Lecture 12

Nonhomogeneous Equations: The Method of Undetermined Coefficients

So far, we have been considering constant coefficient equations that are homogeneous such as

$$ay'' + by' + cy = 0,$$

but not all equations we will want to consider are homogeneous. Given the equation

$$ay'' + by' + cy = f(t)$$

a solution which satisfies the associated homogeneous equation will not satisfy the nonhomogeneous equation unless $f(t) = 0$. Thus, we will need to look for a separate set of solutions to any nonhomogeneous equation.

Definition: A solution, y_p, of the equation

$$ay'' + by' + cy = f(t) \tag{12.1}$$

where $f(t) \neq 0$ is called a **particular solution** if it contains no arbitrary constants.

Example: Given the equation

$$y'' - 4y = -3\sin(t), \quad f(t)$$

show that $y_p(t) = \frac{3}{5}\sin(t)$ is a particular solution.

$y_p' = \frac{3}{5}\cos(t)$, $y_p'' = -\frac{3}{5}\sin(t)$

$y_p'' - 4y_p = \left(-\frac{3}{5}\sin(t)\right) - 4\left(\frac{3}{5}\sin(t)\right) = -\frac{3}{5}\sin(t) - \frac{12}{5}\sin(t)$

$= -\frac{15}{5}\sin(t) = -3\sin(t)$ $f(t)$ ✓

$$ay'' + by' + cy = f(t)$$

y_p ?

Note that, in the previous example $f(t) = -3\sin(t)$ and the particular solution $y_p(t) = \frac{3}{5}\sin(t)$ are of similar forms (both sine functions). In certain circumstances we can use this idea to construct particular solutions by considering the form of the forcing function $f(t)$. This method is called the Method of Undetermined Coefficients.

Method of Undetermined Coefficients:

1. Using the form of $f(t)$ we "guess" a form for the particular solutions $y_p(t)$.
2. Substitute the guessed form into the differential equation.
3. Determine the coefficients that make the guess satisfy the equation

Note: This method only works for functions of certain appropriate forms:

- constant functions (e.g. $f(t) = 10$)
- polynomials (e.g. $f(t) = t^2 + 5$)
- exponentials - $e^{\alpha t}$
- sine or cosine - $\sin(\beta t)$, $\cos(\beta t)$
- any linear combination of the functions above (e.g. $P(t)e^{\alpha t}\sin(\beta t)$ where $P(t)$ is a polynomial and α, β real numbers) $te^t\sin(t)$

forms for a homogeneous solution

$ay'' + by' + cy = 0$

y_h is of this form

This method is not applicable when $f(t) = \ln(t), \frac{1}{t}, \tan(t), \ldots$.

Example: Find a particular solution for

$$y'' + 2y' + 5y = 4t$$

Here $f(t) = 4t$ polynomial of degree 1

Method Coefficients we need to determine them.

① Guess $y_p = \underbrace{At + B}$ constants

$y_p' = A$

$y_p'' = 0$

② $y'' + 2y' + 5y = 0 + 2(A) + 5(At + B) \overset{\text{want}}{=} 4t$

$\Rightarrow \underbrace{5A}t + \underbrace{(2A + 5B)} = \underbrace{4}t + \underbrace{0}$

③ So $\begin{cases} 5A = 4 \\ 2A + 5B = 0 \end{cases}$ $\Rightarrow \boxed{A = \frac{4}{5}}$

$\Rightarrow 2(\frac{4}{5}) + 5B = 0$

$\Rightarrow 5B = -\frac{8}{5}$

$\Rightarrow \boxed{B = -\frac{8}{25}}$

So $\boxed{y_p = \frac{4}{5}t - \frac{8}{25}}$ is the particular solution.

In general, if the differential equation is of the form

$$ay'' + by' + cy = C\underbrace{t^m}$$

t^2

then we can use

$A_2 t^2 + A_1 t + A_0$

$$y_p = \underline{A_m t^m + \cdots + A_1 t + A_0},$$

where $A_0, A_1, \ldots, A_m$ are real numbers, as the form of the particular solution.

Example: Consider the equation

$$y'' + 2y' + 5y = 10e^{3t} \qquad f(t) = 10e^{3t}$$

Guess $y_p = Ce^{3t}$, $y_p' = 3Ce^{3t}$, $y_p'' = 9Ce^{3t}$

$$y'' + 2y' + 5y = 9Ce^{3t} + 2(3Ce^{3t}) + 5(Ce^{3t}) \overset{\text{want}}{=} 10e^{3t}$$

$$\Rightarrow 20Ce^{3t} = 10e^{3t}$$

$$\Rightarrow 20C = 10 \Rightarrow C = \frac{1}{2}$$

Thus $\boxed{y_p = \frac{1}{2}e^{3t}}$

Example: Consider the equation

$$y'' + 2y' + 5y = \sin(t) \qquad f(t) = \sin(t)$$

Try $y_p = A\sin(t)$, $y_p' = A\cos(t)$, $y_p'' = -A\sin(t)$

$$\cancel{y'' + 2y' + 5y = -A\sin(t) + 2A\cos(t) + 5A\sin(t) = \sin(t)}$$

$A = 0$

When we have sine/cosine we need to use both functions in our guess:

$$y_p = A\sin(t) + B\cos(t)$$

$$y_p' = A\cos(t) - B\sin(t), \quad y_p'' = -A\sin(t) - B\cos(t)$$

So $y'' + 2y' + 5y = (-A\sin(t) - B\cos(t)) + 2(A\cos(t) - B\sin(t)) + 5(A\sin(t) + B\cos(t)) \overset{\text{want}}{=} \sin(t)$

$$\Rightarrow (-A - 2B + 5A)\sin(t) + (-B + 2A + 5B)\cos(t) = \sin(t)$$

$\Rightarrow (4A-2B)\sin(t) + (4B+2A)\cos(t) = \sin(t)$
$= 1\cdot\sin(t) + 0\cos(t)$

$$\Rightarrow \begin{cases} 4A-2B=1 \\ 4B+2A=0 \Rightarrow A=-2B \end{cases}$$

$\Rightarrow 4(-2B)-2B=1 \Rightarrow -10B=1 \Rightarrow B=-\frac{1}{10}$
$\Rightarrow A = \frac{2}{10} = \frac{1}{5}$

So $\boxed{y_p = \frac{1}{5}\sin(t) - \frac{1}{10}\cos(t)}$

In general, if the differential equation is of the form

$$ay'' + by' + cy = C\sin(\beta t) \text{ or } (C\cos(\beta t))$$

then we can use

$$y_p = A\cos(\beta t) + B\sin(\beta t),$$

where A and B are real numbers, as the form of the particular solution.

Example: Consider the equation

$$y'' + 4y = 5t^2e^t \qquad f(t) = 5t^2e^t$$

poly exp

Guess: $y_p = (At^2+Bt+C)e^t$

$y_p' = (2At+B)e^t + (At^2+Bt+C)e^t$

$y_p'' = 2Ae^t + (2At+B)e^t + (2At+B)e^t + (At^2+Bt+C)e^t$
$= e^t[2A+2At+B+2At+B+At^2+Bt+C]$
$= e^t[At^2+(4A+B)t+(2A+2B+C)]$

$y''+4y = \cancel{e^t}[At^2+(4A+B)t+(2A+2B+C)] + 4(At^2+Bt+C)\cancel{e^t} = 5t^2\cancel{e^t}$

$\Rightarrow 5At^2 + (4A+5B)t + (2A+2B+5C) = 5t^2+0t+0$

$$\begin{cases} 5A=5 & \Rightarrow \boxed{A=1} \\ 4A+5B=0 & \Rightarrow 4+5B=0 \Rightarrow \boxed{B=-4/5} \\ 2A+2B+5C=0 & \Rightarrow 2+(-\frac{8}{5})+5C=0 \Rightarrow 5C=-\frac{2}{5} \Rightarrow \boxed{C=\frac{-2}{25}} \end{cases}$$

So $\boxed{y_p = (t^2 - \frac{4}{5}t - \frac{2}{25})e^t}$

Example: Consider the equation

$$y'' + y = 10 \quad \leftarrow F(t)$$

poly degree 0

Try use $y_p = C$, $y_p'' = 10$ $\quad 0 + C = 10$

$$y'' + y' = 10$$

Try Guess $y_p = C \Rightarrow y_p' = 0, y_p'' = 0$, substitute. get $\cancel{0 + 0 = 10}$ want

$y = C$ is a solution to the homogeneous eqn. $y'' + y' = 0$

char eqn: $r^2 + r = 0 \Rightarrow r(r+1) = 0$, $r = 0, -1$

$y_h = C + De^{-t}$ $\quad 0 + C = 10 \Rightarrow C = 10$

If repeated we multiply by t. Guess $y_p = C \cdot t$, $y' = C$, $y'' = 0$ $\quad \boxed{y_p = 10t}$

Note: If the form of the particular solution is present in the form of the homogeneous solution, then we must multiply by powers of the independent variable until the form of the particular solution is no longer present in the homogeneous solution.

Guess

Example: Consider the equation

$$y'' - 6y' + 9y = e^{3t}$$

characteristic eqn: $r^2 - 6r + 9 = 0$

$\Rightarrow (r-3)^2 = 0 \Rightarrow r = 3, 3$ repeated

So $y_h = Ce^{3t} + Dte^{3t}$

For $f(t) = e^{3t}$ guess $y_p = Ct^2e^{3t}$ (multiply by t^2 so it is not in homogeneous soln.)

$y_p' = C \cdot 2te^{3t} + Ct^2 \cdot 3e^{3t} = e^{3t}(2Ct + 3Ct^2)$

$y_p'' = 3e^{3t}(2Ct + 3Ct^2) + e^{3t}(2C + 6Ct)$

$= e^{3t}(9Ct^2 + 12Ct + 2C)$

$y'' - 6y' + 9y = \cancel{e^{3t}}(9Ct^2 + 12Ct + 2C) - 6(\cancel{e^{3t}}(2Ct + 3Ct^2))$

$+ 9(\cancel{e^{3t}} C \cdot t^2) \overset{\text{want}}{=} \cancel{e^{3t}} 1$

$(9 - 18 + 9)Ct^2 + (12 - 12)Ct + 2C = 1 \Rightarrow 2C = 1 \Rightarrow C = \frac{1}{2}$ $\quad \boxed{y_p = \frac{1}{2}t^2e^{3t}}$

Given the equation

$$ay'' + by' + cy = f(t),$$

we can use $y_p(t)$ as given in the following table depending upon the form of $f(t)$ and the form of the homogeneous solution.

Table of Particular Solution Forms for the Method of Undetermined Coefficients:

$f(t)$	$y_p(t)$
t^2	$t^s(At^2+Bt+C)$
$P(t)$ polynomial	$t^s Q(t)$, $\deg(Q) = \deg(P)$
$e^{\alpha t}$	$At^s e^{\alpha t}$
t^2e^t	$t^s(At^2+Bt+C)e^t$
$e^{\alpha t}P(t)$	$t^s e^{\alpha t} Q(t)$, $\deg(Q) = \deg(P)$

- $s = 0$ if α is not a root of the characteristic equation.
- $s = 1$ if α is a single root of the characteristic equation.
- $s = 2$ if α is a repeated root of the characteristic equation.

Recall $y_h = Ce^t\sin(t)$

$r = 1 + i$

$\alpha + i\beta$

$f(t)$	$y_p(t)$
$e^{\alpha t} \cdot \begin{cases} \sin(\beta t) \\ \cos(\beta t) \end{cases}$	$t^s e^{\alpha t}(A\cos(\beta t) + B\sin(\beta t))$
$P(t)e^{\alpha t} \cdot \begin{cases} \sin(\beta t) \\ \cos(\beta t) \end{cases}$	$t^s e^{\alpha t}(Q_1(t)\cos(\beta t) + Q_2(t)\sin(\beta t))$

- $s = 0$ if $\alpha + i\beta$ is not a root of the characteristic equation.
- $s = 1$ if $\alpha + i\beta$ is a root of the characteristic equation.

Example: Consider the equation

$$y'' + 2y' + 5y = f(t)$$

Write the form of the particular solution in the table below for each function $f(t)$.

char eqn : $r^2+2r+5=0$

$r = \frac{-2 \pm \sqrt{4-4(1)(5)}}{2} = \frac{-2 \pm \sqrt{-16}}{2}$

$= \frac{-2 \pm 4i}{2} = -1 \pm 2i$ $\quad r = \begin{cases} -1+2i \\ -1-2i \end{cases}$

$f(t)$	$y_p(t)$
$7\cos(3t)$ $\alpha=0, \beta=3$ $r=0+3i$	$A\cos(3t) + B\sin(3t)$
$2e^t$ $r=1$	Ce^t
$2te^t\sin(t)$ $r=1+i$	$(At+B)e^t\sin(t) + (Ct+D)e^t\cos(t)$
$te^{-t}\cos(2t)$ $r=-1+2i$	$t\left[(At+B)e^{-t}\cos(2t) + (Ct+D)e^{-t}\sin(2t)\right]$
t^2e^{-t} $r=-1$	$(At^2+Bt+C)e^{-t}$

Example: Consider the equation

$$y'' - 2y' + y = f(t)$$

Write the form of the particular solution in the table below for each function $f(t)$.

char eqn: $r^2 - 2r + 1 = 0$

$\Rightarrow (r-1)^2 = 0 \Rightarrow r = 1, 1$

$y_h = Ce^t + Dte^t$ twice

$f(t)$	$y_p(t)$
$7\cos(3t)$ $r = 0 + 3i$	$A\cos(3t) + B\sin(3t)$
$2te^t\sin(t)$ $r = 1 + i$	$(At+B)e^t\sin(t) + (Ct+D)e^t\cos(t)$
$t^2\boxed{e^t}$ $r = 1$	$\underline{t}^2\left[\underbrace{(At^2+Bt+C)e^t}_{\text{normal guess}}\right]$

This method can be used for higher order constant coefficient equations.

Example: Consider the equation

$$y''' + 2y'' + y' = \frac{\cos(2t)}{e^t} = \underbrace{e^{-t}\cos(2t)}_{r = -1 + 2i}$$

What is the form of the particular solution?

char eqn: $r^3 + 2r^2 + r = 0$

$r(r^2 + 2r + 1) = 0$

$r(r+1)^2 = 0, \quad r = 0, -1, -1$

$y_h = \underset{r=0}{C_1} + \underbrace{C_2e^{-t} + C_3te^{-t}}_{r=-1}$

use $\boxed{y_p = Ae^{-t}\cos(2t) + Be^{-t}\sin(2t)}$

Example: Can we use the method of undetermined coefficients to find particular solutions to the following equations?

e^t $\quad f(t) = e^{\sim} \sin(\sim)$ $\quad P(t)$

- $y'' + 4y = 2^t \sin(3t)$

$\to 2^t \sin(3t)$

$= (e^{\ln(2)})^t \sin(3t)$

$e^{\ln(2)t} \sin(3t)$ ✓

$2 = e^{\ln(2)}$

- $y'' + ty' - t^2 y = \sin(t)$ Good for $f(t)$

variable coefficients ✗ can't use the method!